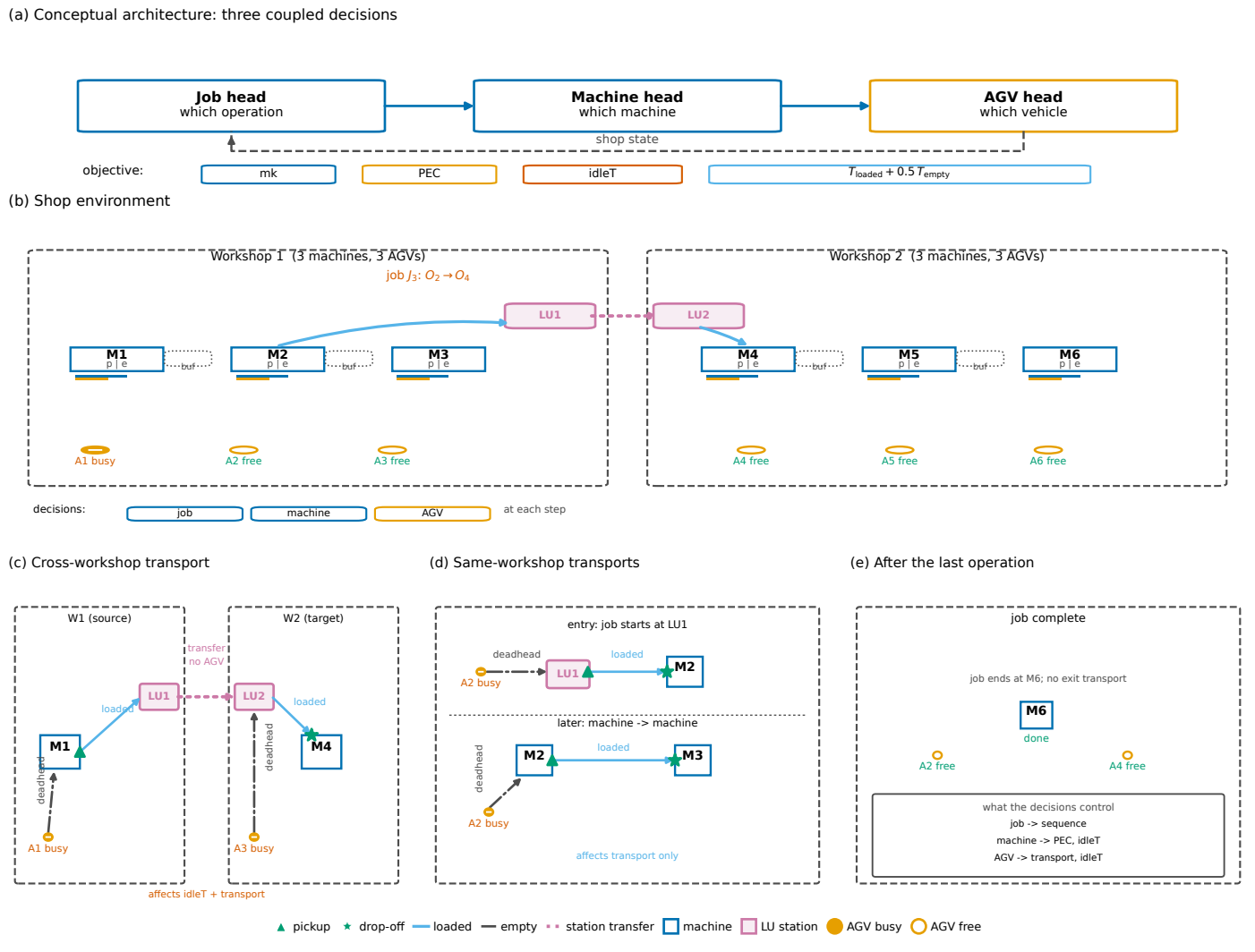

(a) Conceptual architecture: three coupled decisions
Job head
which operation
Machine head
which machine
AGV head
which vehicle
shop state
objective:
mk
PEC
idleT
$T_{\text{loaded}} + 0.5\,T_{\text{empty}}$
(b) Shop environment
Workshop 1 (3 machines, 3 AGVs)
job J_3: $O_2 \rightarrow O_4$
Workshop 2 (3 machines, 3 AGVs)
LU1
LU2
M1
M2
M3
M4
M5
M6
p | e
buf
A1 busy
A2 free
A3 free
A4 free
A5 free
A6 free
decisions:
job
machine
AGV
at each step
(c) Cross-workshop transport
W1 (source)
W2 (target)
transfer
no AGV
loaded
deadhead
A1 busy
A3 busy
affects idleT + transport
(d) Same-workshop transports
entry: job starts at LU1
A2 busy
later: machine -> machine
affects transport only
(e) After the last operation
job complete
job ends at M6; no exit transport
done
what the decisions control
job -> sequence
machine -> PEC, idleT
AGV -> transport, idleT
pickup
drop-off
loaded
empty
station transfer
machine
LU station
AGV busy
AGV free